PROJECT REPORT

Predicting Task Accuracy from Digital Multitasking Behaviour Using Statistical Techniques

(A Statistical Study of Attention and Digital Interruptions)

A PROJECT SUBMITTED TO

DEPARTMENT OF STATISTICS

BHAVAN'S COLLEGE (AUTONOMOUS), ANDHERI (W)

Master of Science in Applied Statistics, Part -1

SEMESTER II

Submitted by,

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MSAS116

UNDER THE GUIDANCE OF

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CERTIFICATE

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This is to certify that, Shaikh Maryam Alimuddin has successfully completed the Project work entitled "Predicting Task Accuracy from Digital Multitasking Behaviour Using Statistical Techniques" in M.Sc. (Applied Statistics) Part I under my supervision in this college, during the Semester II of academic year 2025-26.

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Acknowledgement

I sincerely thank **Ms. Ruheeen Qureshi** for her expert guidance, valuable feedback, and continuous support throughout the completion of this project. Her insights and suggestions were instrumental in shaping the methodology and ensuring the academic rigor of this study.

I would also like to express my sincere gratitude to **Prof. Rakhi Khambale, Head of the Department of Statistics**, for her encouragement, support, and for providing an academic environment that fostered learning and research.

I am grateful to the **Department of Statistics, Bhavan's College**, for providing an enriching learning environment and the necessary resources that made this research possible.

I would also like to thank all the participants who took the time to complete the survey and contribute their responses, which formed the foundation of this study.

Special appreciation goes to my classmates and peers for their constructive discussions and helpful suggestions, which helped refine the clarity and depth of this work.

Finally, I extend my heartfelt gratitude to my **family and friends** for their constant encouragement, understanding, and patience, which kept me motivated throughout the research and writing process.

Shaikh Maryam
17 March 2026

Abstract

The rapid expansion of digital technologies has significantly increased the prevalence of media multitasking among students. Modern learning environments frequently involve simultaneous interaction with multiple digital streams such as smartphones, laptops, notifications, and background media. While digital tools provide flexibility and access to information, they may also introduce attentional disruptions that influence cognitive performance. Previous research, including the study by Uncapher, Thieu, and Wagner (2016), suggests that heavy media multitaskers may exhibit reduced attentional control and impaired memory performance. However, much of the existing literature relies on controlled laboratory experiments that focus primarily on memory tasks rather than real-world academic task performance.

The present study extends this research by examining the relationship between digital multitasking behaviour and cognitive task accuracy in a realistic student learning context. Data were collected through a cross-sectional survey administered using Google Forms, combined with a 15-question cognitive reasoning test designed to measure task accuracy. The final dataset consisted of responses from 139 students, including demographic variables and behavioural indicators related to digital study habits. Key predictors included screen time, notification frequency, and self-reported focus level. A composite Multitasking Index was constructed by combining four behavioural indicators: app switching, phone checking during study sessions, background media usage, and multi-device usage.

The research workflow involved structured data storage, SQL-based data cleaning, feature engineering, exploratory data analysis, and statistical modelling. Analytical methods included descriptive statistics, correlation analysis, multiple linear regression, interaction models, relative importance analysis, clustering analysis, ANOVA, mediation analysis, and regression diagnostic checks.

The results indicate that multitasking behaviour alone does not significantly predict task accuracy. Instead, notification frequency shows a significant negative relationship with task performance, while higher focus levels are associated with improved cognitive accuracy. Interaction analysis further suggests that multitasking becomes more disruptive when students experience frequent digital interruptions. Mediation analysis also indicates that multitasking behaviour indirectly influences performance through attentional mechanisms such as increased notifications and reduced focus.

Overall, the findings suggest digital interruptions and attentional control play a more important role in determining cognitive task performance than multitasking behaviour itself. These results highlight the importance of managing digital distractions in modern educational environments and contribute to the growing literature on digital multitasking and cognitive performance.

Keywords:
Digital Multitasking, Cognitive Task Accuracy, Notification Interruptions, Attentional Focus, Multiple Linear Regression, Behavioral Data Analysis, Student Study Behaviour

1. Introduction

1.1 Digital Multitasking in Contemporary Learning Environments

The rapid expansion of digital technologies has fundamentally reshaped modern learning environments. Students today frequently interact with multiple digital platforms simultaneously, including smartphones, laptops, messaging applications, online learning systems, and background media. This behaviour, commonly referred to as **digital or media multitasking**, involves switching attention between multiple streams of information within short intervals. While digital technologies provide convenient access to information and learning resources, they also introduce frequent interruptions that may influence cognitive processes such as attention, reasoning, and task performance.

A growing body of research has investigated the cognitive consequences of media multitasking. In particular, Uncapher, Thieu, and Wagner (2016) examined how media multitasking behaviour relates to cognitive control and memory performance. Their findings suggest that individuals who frequently engage in media multitasking often demonstrate reduced attentional control and increased susceptibility to distraction, making it more difficult to filter irrelevant information. These attentional difficulties may impair cognitive processes such as working memory, information processing, and decision-making.

Despite these insights, much of the existing research relies on laboratory-based experiments that primarily examine memory performance under controlled conditions. Although such experimental designs provide valuable insights into cognitive mechanisms, they may not fully capture the complexity of real-world learning environments where students are exposed to continuous digital interruptions, device switching, and multitasking behaviour during academic activities.

1.2 Behavioural Complexity of Digital Study Habits

Modern students operate within digitally intensive environments characterized by constant connectivity and multiple competing information streams. Notifications from messaging applications, social media platforms, and academic tools can interrupt ongoing tasks and fragment attention. In addition, behaviours such as frequent phone checking, app switching, and simultaneous use of multiple devices may further increase cognitive load and reduce sustained concentration.

To capture these behavioural patterns, the present study collects several indicators of digital multitasking behaviour among students. The dataset consists of responses from **139 participants**, each completing a **15-question cognitive reasoning test** designed to measure task accuracy. Alongside cognitive performance measures, behavioural variables describing digital study habits were collected, including screen time, notification frequency, and self-reported focus level during task completion.

A composite **Multitasking Index** was constructed by combining four behavioural indicators: app switching behaviour, phone checking during study sessions, background media consumption, and simultaneous use of multiple devices. This index provides a quantitative measure of overall multitasking intensity, allowing statistical evaluation of how digital multitasking relates to cognitive task accuracy.

1.3 Study Objectives

The present study aims to examine the relationship between digital multitasking behaviour and cognitive task accuracy among students in a realistic digital learning environment. The specific objectives of the study are:

• To examine whether digital multitasking behaviour predicts task accuracy among students.

• To evaluate the effect of notification frequency on cognitive task performance.

• To assess the relationship between attentional focus and task accuracy.

• To investigate whether multitasking behaviour becomes more disruptive when digital interruptions increase.

• To identify behavioural patterns among students based on their digital study habits.

1.4 Statistical Perspective on Behavioural Data

From a statistical perspective, analysing behavioural data related to digital study habits requires careful modelling of multiple interacting factors. Behavioural datasets often include correlated predictors, measurement variability, and potential indirect effects that complicate simple causal interpretation. For example, multitasking behaviour may not directly influence task accuracy but may instead operate through intermediate mechanisms such as increased interruptions or reduced attentional focus.

To address these complexities, the present study employs a structured statistical framework. The analysis begins with exploratory data analysis and descriptive statistics to examine the distribution of behavioural and performance variables. Correlation analysis is then used to identify preliminary relationships among predictors.

Subsequently, multiple regression models are estimated to evaluate whether multitasking behaviour predicts task accuracy while controlling for additional behavioural variables such as notification frequency, screen time, and focus level. Interaction models are also considered to investigate whether the impact of multitasking behaviour depends on the frequency of digital interruptions. Additional analytical techniques—including clustering analysis and mediation modelling—are used to explore behavioural mechanisms that may explain observed relationships in the data.

1.5 Research Gap and Study Contribution

Although previous studies have provided important insights into the cognitive effects of media multitasking, several limitations remain. Many existing investigations focus primarily on memory tasks conducted in controlled laboratory settings rather than examining behavioural data from real student learning environments. Furthermore, relatively few studies investigate how modern digital behaviours—such as frequent notifications, multi-device usage, and continuous connectivity—interact with multitasking behaviour to influence cognitive task performance.

The present study addresses these limitations by analysing real behavioural data collected from students engaged in a reasoning-based cognitive task. By integrating behavioural indicators of digital study habits with statistical modelling techniques, the research provides a more comprehensive examination of how digital multitasking, attentional focus, and notification interruptions interact to influence task accuracy.

1.6 Research Objectives and Hypotheses

The primary objective of this study is to investigate whether digital multitasking behaviour influences cognitive task accuracy among students. In addition, the study examines how attentional factors such as notification interruptions and focus levels contribute to task performance.

The specific objectives of this research are:

1. Examine whether digital multitasking behaviour predicts task accuracy.
2. Evaluate the effect of notification frequency on cognitive task performance.
3. Assess the relationship between attentional focus and task accuracy.
4. Investigate whether multitasking behaviour becomes more disruptive when digital interruptions increase.
5. Identify behavioural patterns among students using clustering techniques.

To evaluate these objectives, the following statistical hypotheses are formulated:

H_{01}: Digital multitasking behaviour has no significant effect on task accuracy.
H_{02}: Notification frequency has no significant effect on task accuracy.
H_{03}: Focus level has no significant effect on task accuracy.
H_{04}: There is no interaction effect between multitasking behaviour and notification frequency in predicting task accuracy.

Rejection of these hypotheses would suggest that digital interruptions and attentional mechanisms play a meaningful role in determining cognitive task performance in modern digital learning environments.

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2. Literature Review

2.1 Digital Media Multitasking

The rapid expansion of digital technology in the 21st century has significantly increased the prevalence of digital media multitasking. Digital media multitasking refers to the simultaneous engagement with multiple digital media streams, such as smartphones, laptops, social media platforms, and messaging applications. Among students, this behaviour has become an integral part of both academic and everyday activities, often involving frequent switching between educational tasks and non-academic digital interactions. Rideout, Foehr, and Roberts (2010) reported that adolescents and young adults engage in media multitasking during a substantial portion of their daily media consumption, raising concerns regarding its potential cognitive consequences.

Early research conceptualized media multitasking as a behavioural tendency rather than a cognitive skill. In other words, individuals differ in their propensity to engage in simultaneous media use rather than in their ability to effectively manage multiple information streams. Ophir, Nass, and Wagner (2009) introduced the **Media Multitasking Index (MMI)** as a quantitative measure of multitasking behaviour and demonstrated that individuals who frequently engage in multiple forms of media simultaneously—referred to as heavy media multitaskers (HMMs)—exhibit distinct attentional characteristics compared to light media multitaskers (LMMs). Their findings suggested that heavy multitaskers may experience greater difficulty filtering irrelevant information, which can influence cognitive performance in attention-demanding tasks.

These findings initiated a growing body of research examining how digital multitasking behaviour relates to cognitive functioning, particularly in environments where individuals must continuously manage competing sources of information.

2.2 Media Multitasking and Cognitive Performance

A substantial portion of existing research has focused on the relationship between media multitasking and cognitive processes such as working memory, attention, and cognitive control. Uncapher, Thieu, and Wagner (2016) provided strong evidence that chronic media multitasking is associated with reduced working memory capacity and poorer long-term memory performance. Their results indicate that heavy media multitaskers tend to store information less precisely, which may lead to difficulties in both short-term information maintenance and long-term retrieval.

Other studies have further demonstrated that heavy media multitaskers often experience difficulties in filtering distracting stimuli. Cain and Mitroff (2011) showed that individuals who frequently engage in media multitasking are more vulnerable to distraction, even when instructed to focus on task-relevant information. This reduced ability to filter irrelevant stimuli may negatively affect task performance, particularly in environments characterized by competing digital inputs.

Research examining task-switching abilities among media multitaskers has produced mixed findings. Some studies report that heavy media multitaskers perform worse on task-switching and attentional control tasks (Ophir et al., 2009; Sanbonmatsu et al., 2013). In contrast, other studies suggest that multitaskers may perform similarly—or occasionally better—than non-multitaskers in certain contexts (Alzahabi and Becker, 2013). These inconsistent results suggest that digital multitasking may not uniformly impair cognitive functioning but may instead influence specific processes related to attentional control and information-processing precision.

2.3 Impulsivity, Attention, and Multitasking Behaviour

Another important psychological factor associated with digital multitasking behaviour is impulsivity, particularly **attentional impulsivity**. Several studies have reported positive associations between media multitasking behaviour and impulsivity-related traits (Minear et al., 2013; Sanbonmatsu et al., 2013). Individuals who frequently engage in media multitasking often report greater difficulty sustaining attention and are more susceptible to intrusive or distracting thoughts.

Uncapher et al. (2016) further demonstrated that attentional impulsivity is specifically linked to reduced working memory discrimination rather than careless decision-making. This suggests that the cognitive cost of multitasking may arise from reduced precision in information processing rather than from biased response strategies. In other words, multitasking may degrade the quality of stored information rather than directly influencing decision accuracy.

Despite the growing evidence linking multitasking, attentional impulsivity, and cognitive performance, the direction of causality remains unclear. It is not yet known whether frequent digital multitasking contributes to attentional deficits or whether individuals with pre-existing attentional characteristics are more likely to engage in multitasking behaviour. Nevertheless, the consistent association between multitasking, attentional control, and cognitive performance highlights the importance of further investigation, particularly within student populations.

2.4 Research Gap and Rationale for the Present Study

Although prior research has provided valuable insights into the cognitive consequences of media multitasking, several limitations remain. Much of the existing literature focuses on laboratory-based experiments designed to measure working memory or attentional control under controlled experimental conditions. While these studies provide important theoretical insights, they may not fully capture the complexity of real-world digital learning environments. In particular, relatively few studies have examined how digital multitasking influences **task accuracy in realistic online settings**, where students frequently interact with multiple digital devices, receive notifications, and engage in background media consumption during academic tasks.

Task accuracy represents a practical and performance-based outcome that reflects real-world academic demands more directly than memory recall tasks alone. With the increasing

integration of digital technologies into educational environments—including online learning platforms, digital assessments, and AI-assisted tools—it has become increasingly important to understand how multitasking behaviour affects students' ability to perform cognitive tasks accurately.

The present study addresses these limitations by extending the framework proposed by Uncapher et al. (2016) to an online behavioural context. Instead of laboratory-based memory paradigms, this research employs a **timed task accuracy test** designed to measure reasoning, attention, and pattern recognition abilities in a realistic digital environment. By collecting primary data through a cross-sectional survey and applying statistical modelling techniques, this study aims to investigate whether higher levels of digital multitasking behaviour are associated with reduced task accuracy among students.

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3. Methodology

3.1 Research Design

This study employs a quantitative, cross-sectional research design to investigate the relationship between digital multitasking behaviour and cognitive task accuracy among students. The research integrates survey-based behavioural data with a short cognitive reasoning assessment to examine how digital study habits influence task performance.

A cross-sectional design was selected because it allows the simultaneous observation of behavioural variables and cognitive outcomes within a single data collection period. This approach is widely used in behavioural and educational research when the objective is to examine statistical relationships between lifestyle behaviours and performance outcomes.

The analytical framework of the study combines exploratory data analysis with inferential statistical modelling. The primary goal is to determine whether digital multitasking behaviour, notification interruptions, and attentional focus significantly influence task accuracy.

3.2 Data Collection Procedure

Data were collected through an online survey administered using Google Forms. Participants were invited to complete a questionnaire designed to capture digital study habits, followed by a short cognitive task test.

The survey instrument consisted of three main components:

1. **Demographic Information**
   Participants reported demographic characteristics including age group and gender.

2. **Digital Behaviour Indicators**
   Participants responded to questions describing their digital study habits, including:
   - frequency of app switching during study sessions
   - phone checking behaviour
   - use of multiple devices simultaneously
   - background media consumption while studying
   - daily screen time
   - notification frequency during study sessions

3. **Cognitive Task Assessment**
   Participants completed a 15-question reasoning test designed to evaluate task accuracy. The test included questions measuring logical reasoning, pattern recognition, and attention to detail.

A total of **139 valid responses** were obtained. Each observation in the dataset therefore contains both behavioural indicators and cognitive task performance measures.

Participants were recruited using a **convenience sampling approach**, where students voluntarily completed the online survey.

3.3 Data Processing

The collected survey data were processed through a structured preprocessing pipeline to ensure accuracy, consistency, and suitability for statistical analysis. The data processing procedure included dataset storage, data cleaning, variable transformation, handling of missing values, and feature engineering. These steps ensured that the dataset was analytically reliable and reproducible for subsequent statistical modelling.

3.3.1 Data Storage and Dataset Preparation

Survey responses were collected using Google Forms, which automatically compiled responses into a spreadsheet format. The raw dataset was exported as a CSV file and subsequently imported into a MySQL database to facilitate structured data management and reproducible preprocessing.

Using a relational database environment enabled efficient manipulation of variables, systematic tracking of transformations, and improved data integrity during the cleaning process. This workflow ensured that the dataset could be consistently reproduced and reprocessed if necessary.

3.3.2 Data Cleaning and Validation

Initial inspection of the raw dataset revealed several formatting inconsistencies and spreadsheet-induced errors that required correction before statistical analysis.

A primary issue occurred in the **score variable**, which represented the number of correct answers in the 15-question cognitive test. Due to spreadsheet auto-formatting, some values such as *10/16* and *11/16* were automatically converted into date formats (e.g., *Oct-16* and *Nov-16*). These entries were manually verified and corrected to restore their intended numeric values.

The corrected scores were stored in a cleaned numeric variable (*score_clean*) representing the participant's total number of correct responses. The resulting variable ranged from **0 to 15**, reflecting task accuracy.

Additional cleaning procedures included:

• checking for inconsistent formatting across observations
• verifying categorical responses across participants
• validating demographic variables
• correcting spreadsheet-induced formatting errors

These steps ensured that the dataset accurately reflected the collected responses and minimized potential preprocessing bias.

3.3.3 Handling of Missing Data

The dataset was screened for missing or incomplete responses prior to analysis. Records with incomplete or invalid entries in key variables were reviewed and corrected where possible. Observations with incomplete cognitive test responses were excluded to maintain consistency in the dependent variable.

After preprocessing, the final dataset contained **139 valid observations**, which were used for subsequent statistical analysis.

3.3.4 Variable Encoding

Several survey responses were originally recorded as categorical or ordinal text responses. To enable statistical analysis, these variables were converted into numerical codes.

Notification Frequency

Participants reported notification frequency during study sessions using categorical ranges. These responses were encoded as ordinal numeric values:

Original Response	Code
0–5	1
6–10	2
11–20	3
More than 20	4

This variable was stored as **notifications_code**.

Screen Time

Daily screen exposure during study periods was measured using categorical time ranges and recoded into ordinal values:

Original Response	Code
Below 2 hours	1
2–4 hours	2
4–6 hours	3
Above 6 hours	4

The encoded variable was stored as **screen_time_code**.

Multitasking Behaviour Variables

Three behavioural variables measuring multitasking frequency were recorded using a Likert-type scale and encoded numerically as follows:

Response	Code
Never	0
Rarely	1
Sometimes	2
Often	3
Always	4

These variables included:

• app switching while studying
• phone checking during study sessions
• background media usage while studying

In addition, **multi-device usage** was measured as a binary variable indicating whether participants used more than one device simultaneously during study sessions:

Response	Code
No	0
Yes	1

These transformations enabled quantitative analysis of multitasking behaviour and ensured compatibility with statistical modelling techniques.

3.3.5 Feature Engineering: Multitasking Index

To represent overall multitasking behaviour during study sessions, a composite variable called the **Multitasking Index** was constructed. The index integrates four behavioural indicators collected in the survey:

• app switching while studying
• phone checking behaviour
• background media consumption
• multi-device usage

The index was calculated as the average of the four behavioural variables:
Multitasking Index =
(AppSwitching + PhoneChecking + MultiDeviceUse + BackgroundMedia) / 4

Each component contributes equally to the index, allowing it to capture multiple dimensions of digital multitasking behaviour.

Because the component variables have different measurement ranges, the resulting Multitasking Index ranges approximately from **0 to 3**, where lower values indicate minimal multitasking behaviour and higher values represent more intensive multitasking activity.

The multitasking index was used as the **primary independent variable** in subsequent statistical analyses examining the relationship between digital multitasking behaviour and cognitive task accuracy.

3.3.6 Software Used

Data cleaning, preprocessing, and feature construction were performed using **MySQL** for structured data management and **R** for statistical preprocessing and dataset preparation. These tools enabled reproducible data processing and ensured compatibility with subsequent statistical analysis procedures.

3.3.7 Final Analysis Dataset

After preprocessing and feature construction, the final dataset consisted of **139 observations** and included the following variables used for statistical analysis:

Variable	Description	Role
score	Cognitive task accuracy	Dependent variable
multitasking_index	Composite multitasking behaviour measure	Main predictor
screen_time_code	Screen exposure during study	Control variable
notifications_code	Notification frequency	Control variable
focus_level	Self-reported concentration level	Behavioural predictor
gender	Participant gender	Demographic variable
age_group	Participant age category	Demographic variable

The cleaned and processed dataset was then exported for statistical analysis and modelling.

3.4 Variable Definition and Measurement

The study includes one primary outcome variable along with several behavioural predictor variables.

Dependent Variable

The dependent variable in this study is **task accuracy**, measured as the number of correctly answered questions in the cognitive test.

Score Range:
0 – 15

Higher scores indicate better cognitive task performance.

Independent Variables

Several behavioural predictors were included in the analysis:

• **Multitasking Index** – Composite measure representing overall digital multitasking behaviour.
• **Notification Frequency** – Frequency of digital notifications received during study sessions.
• **Screen Time** – Daily screen exposure category.
• **Focus Level** – Self-reported attentional focus while performing tasks.
These variables capture key aspects of students' digital study environments.

Demographic Variables

To account for potential differences across participants, demographic variables were also recorded:

• Age group
• Gender

3.5 Statistical Analysis Framework

The statistical analysis was conducted in several stages.

Descriptive Statistics

Initial descriptive statistics were computed to summarize the dataset. Measures included mean, standard deviation, minimum, and maximum values for the primary behavioural variables and task accuracy scores.

Exploratory Data Analysis

Exploratory data analysis (EDA) was performed to visualize the distribution of variables and identify potential relationships within the dataset. Visualizations included histograms, scatterplots, boxplots, and correlation heatmaps.

Correlation Analysis

Pearson correlation coefficients were calculated to evaluate pairwise relationships between multitasking behaviour, notification frequency, screen time, focus level, and task accuracy.

3.6 Overview of Statistical Analysis

The statistical analysis in this study was conducted in several stages to systematically investigate the relationship between digital multitasking behaviour and cognitive task accuracy. The analysis combined descriptive statistics, inferential modelling, diagnostic testing, and advanced behavioural modelling techniques. In total, approximately **twenty statistical procedures and analytical techniques** were applied to examine the dataset.

3.6.1 Descriptive Statistics

Descriptive statistics were first calculated to summarize the basic characteristics of the dataset. These measures included the **mean, standard deviation, minimum, and maximum values** for the primary variables.

Descriptive statistics provide an overview of the distribution of the variables and help identify general patterns within the dataset. For example, the average task accuracy score in the dataset was approximately **9.71**, indicating moderate performance among participants.

3.6.2 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was conducted to visually examine the distribution of variables and identify potential relationships between behavioural predictors and task accuracy.

Graphical techniques used in this stage included:

• histograms to examine score distributions
• scatterplots to inspect relationships between multitasking behaviour and task performance
• distribution plots for behavioural variables

EDA provides an intuitive understanding of the dataset before formal statistical modelling is applied.

3.6.3 Correlation Analysis

Pearson correlation coefficients were computed to evaluate the strength and direction of relationships between behavioural variables and cognitive task accuracy.
Key correlations examined included:

• score vs multitasking behaviour
• score vs notification frequency
• score vs focus level
• score vs screen time

For example, the correlation between **focus level and task accuracy (r = 0.23)** suggested a positive relationship between attentional focus and performance.
Correlation analysis provided an initial indication of potential predictors before regression modelling was conducted.

Assumptions of Pearson Correlation

Pearson correlation analysis relies on several assumptions:

• **Linearity** – The relationship between the two variables should be approximately linear. This assumption was examined using scatterplots, which indicated no strong deviations from linearity.

• **Normality** – The variables should be approximately normally distributed. The distribution of task scores appeared reasonably symmetric, and the relatively large sample size (n = 139) supports the use of Pearson correlation.

• **Independence of observations** – Each observation should represent an independent participant.

Since survey responses were collected from different individuals, this assumption was satisfied.

3.6.4 Regression-Based Analyses

Regression modelling formed the core inferential framework of the study. Regression analysis is a statistical technique used to examine how one or more independent variables influence a continuous dependent variable. It allows estimation of the direction and magnitude of relationships between predictors and outcomes while controlling for other variables.

In this study, regression models were used to evaluate how behavioural variables related to digital study habits influence cognitive task accuracy.

Simple Linear Regression

A simple linear regression model was first estimated to examine whether multitasking behaviour alone predicts task accuracy.

Model:
Score ~ MultitaskingIndex

This model provides a baseline assessment of the relationship between multitasking behaviour and cognitive performance. Simple regression estimates the expected change in task accuracy associated with a one-unit change in multitasking behaviour.

The model can be expressed as:

$$Score = \beta_0 + \beta_1(MultitaskingIndex) + \varepsilon$$

where
β_0 represents the intercept,
β_1 represents the regression coefficient for multitasking behaviour,
and ε represents the random error term.

Assumptions of Simple Linear Regression
The validity of linear regression results depends on several key assumptions:

1. Linearity
The relationship between the predictor and the outcome variable should be linear.
This assumption was assessed through residual plots. The residuals were randomly scattered around zero, indicating that the linear model was appropriate.

2. Independence of Errors
Observations should be independent and residuals should not be correlated.
Since the data were collected from different participants responding independently to the survey, the independence assumption is reasonably satisfied.

3. Homoscedasticity
The variance of residuals should remain constant across all levels of the predictor variable.
This assumption was evaluated using residual vs fitted plots. The spread of residuals appeared relatively constant, suggesting no major heteroscedasticity.

4. Normality of Residuals
Residuals should be approximately normally distributed.
Normal Q-Q plots were examined and indicated that residuals followed an approximately normal distribution.

5. Absence of Extreme Outliers
Extreme observations may disproportionately influence regression estimates.
Cook's distance was examined to detect influential observations. No observations exceeded commonly used thresholds, indicating that no individual cases exerted excessive influence on the model.

Multiple Linear Regression

A more comprehensive model was then estimated including additional behavioural predictors:

Score ~ MultitaskingIndex + ScreenTime + Notifications + FocusLevel

Multiple regression allows simultaneous evaluation of several predictors influencing task performance. This approach controls for confounding effects and provides a clearer estimate of the independent contribution of each behavioural variable.

The general form of the model is:
$Score = \beta_0 + \beta_1(MultitaskingIndex) + \beta_2(ScreenTime) + \beta_3(Notifications) + \beta_4(FocusLevel) + \varepsilon$

This model evaluates whether multitasking behaviour, screen time, notification frequency, and attentional focus collectively influence task accuracy.

Assumptions of Multiple Linear Regression

In addition to the assumptions described for simple regression, multiple regression requires the following:

Linearity
Each predictor variable should have a linear relationship with the dependent variable. Residual plots suggested no strong deviations from linearity.

Independence of Errors
Responses were obtained from independent participants, supporting the assumption of independent observations.

Homoscedasticity
Residual plots indicated approximately constant variance of residuals across fitted values.

Normality of Residuals
Normal Q-Q plots indicated approximate normality of the residual distribution.

Absence of Multicollinearity
Predictor variables should not be highly correlated with each other.

Multicollinearity was assessed using the **Variance Inflation Factor (VIF)**.
Observed VIF values ranged between approximately **1.06 and 1.23**, which are well below commonly accepted thresholds (VIF < 5). This indicates that multicollinearity was not a concern in the regression models.

Interaction Regression Model

To examine whether multitasking behaviour becomes more disruptive in the presence of digital interruptions, an interaction term was included:

Score ~ MultitaskingIndex + Notifications + MultitaskingIndex × Notifications + ScreenTime + FocusLevel

Interaction terms allow the effect of one predictor to depend on the level of another predictor. The model is expressed as:

$$\text{Score} = \beta_0 + \beta_1(\text{MultitaskingIndex}) + \beta_2(\text{Notifications}) + \beta_3(\text{MultitaskingIndex} \times \text{Notifications}) + \beta_4(\text{ScreenTime}) + \beta_5(\text{FocusLevel}) + \varepsilon$$

This model tests whether the impact of multitasking behaviour varies depending on the frequency of notifications received during study sessions.

Assumptions of Interaction Models

Interaction regression models follow the same assumptions as multiple linear regression, including:

- linearity
- independence of errors
- homoscedasticity
- normality of residuals
- absence of multicollinearity

These assumptions were evaluated using residual diagnostic plots and VIF statistics. The diagnostics indicated that the regression assumptions were reasonably satisfied.

Focus Interaction Model

An additional interaction regression model was estimated to examine whether attentional focus moderates the effect of multitasking behaviour:

Score ~ MultitaskingIndex × FocusLevel

The corresponding model is:
$Score = \beta_0 + \beta_1(MultitaskingIndex) + \beta_2(FocusLevel) + \beta_3(MultitaskingIndex \times FocusLevel) + \varepsilon$
This model evaluates whether students with higher focus levels experience reduced negative effects of multitasking behaviour.

Regression Model Including Completion Time

A further regression model included task completion time as a predictor variable to investigate the relationship between speed and accuracy.

This model allows examination of a potential **speed–accuracy tradeoff**, where individuals who spend more time on the task may achieve higher accuracy.

The same regression assumptions described above apply to this model and were assessed using diagnostic plots.

Assumption Diagnostics Summary

To ensure the reliability of the regression analyses, several diagnostic procedures were conducted:

• **Residual plots** confirmed approximate linearity and homoscedasticity
• **Normal Q-Q plots** indicated approximate normality of residuals
• **Variance Inflation Factor (VIF)** values between **1.06–1.23** suggested no multicollinearity
• **Cook's distance** analysis showed no highly influential observations

These results indicate that the regression assumptions were reasonably satisfied, supporting the validity of the regression-based statistical inferences.

3.6.5 Additional Statistical Analyses

Relative Importance Analysis

Relative importance analysis was conducted to determine the contribution of each predictor variable to the overall explanatory power of the regression model.

The **LMG method** (Lindeman, Merenda, and Gold) was used to decompose the model's coefficient of determination (R^2) into portions attributable to each predictor.

This approach provides a more interpretable assessment of predictor importance compared to raw regression coefficients because it accounts for correlations between predictors.

Relative importance values were computed for:

- Multitasking Index
- Screen Time
- Notification Frequency
- Focus Level
- Interaction effects

This analysis helps identify which behavioural variables contribute most strongly to explaining variation in task accuracy.

Assumptions of Relative Importance Analysis

Relative importance analysis is based on the fitted multiple regression model and therefore inherits the assumptions of linear regression:
• linear relationships between predictors and outcome
• independence of observations
• homoscedasticity of residuals
• approximate normality of residuals
• absence of severe multicollinearity

Since the regression diagnostics confirmed that these assumptions were reasonably satisfied, the relative importance analysis results are considered reliable.

K-Means Clustering Analysis

To explore behavioural patterns among participants, **K-means clustering** was applied using the following variables:

- Multitasking index
- Notification frequency
- Screen time
- Focus level

Clustering is an **unsupervised machine learning technique** that partitions observations into groups based on similarity across multiple variables.

Before clustering, all variables were standardized to ensure equal contribution to distance calculations.

The optimal number of clusters was determined using the **elbow method**, which evaluates the reduction in within-cluster sum of squares as the number of clusters increases.

The clustering analysis aimed to identify distinct patterns of digital study behaviour among students.

Assumptions of K-Means Clustering

K-means clustering relies on several assumptions:
• **Numeric variables** – All clustering variables must be numeric. This condition was satisfied through prior variable encoding.
• **Comparable variable scales** – Variables should be standardized so that no variable dominates distance calculations. All clustering variables were standardized prior to analysis.
• **Cluster structure** – The data should contain meaningful group structures. The elbow method suggested that a three-cluster solution provided a reasonable balance between model simplicity and explanatory power.
These conditions support the use of K-means clustering to identify behavioural patterns among students.

Mediation Analysis

Mediation analysis was conducted to examine whether the relationship between multitasking behaviour and task accuracy operates indirectly through attentional mechanisms.

Two mediation models were estimated:

Model 1
Multitasking → Focus → Score

Model 2
Multitasking → Notifications → Score

Bootstrapping procedures were used to estimate the **Average Causal Mediation Effect (ACME)** and the **Average Direct Effect (ADE)**.

Mediation analysis allows identification of indirect pathways through which behavioural variables influence cognitive task performance.

Assumptions of Mediation Analysis

Mediation analysis assumes that:
• the independent variable influences the mediator
• the mediator influences the outcome variable
• the regression relationships are correctly specified
• residuals are approximately independent and normally distributed

Bootstrapping procedures were used to estimate indirect effects and confidence intervals, which reduces reliance on strict normality assumptions. The use of bootstrapping therefore strengthens the robustness of the mediation results.

Speed–Accuracy Relationship

The relationship between task completion time and task accuracy was investigated to examine a potential **speed–accuracy tradeoff**.

Both correlation and regression analyses were used to evaluate whether longer task completion times were associated with improved accuracy.

This analysis helps determine whether participants who spend more time on cognitive tasks achieve higher performance.

Assumptions of Speed–Accuracy Analysis

The correlation and regression analyses used to examine the speed–accuracy relationship assume:
• a linear relationship between completion time and task accuracy
• independence of observations
• approximate normality of residuals in regression models

Diagnostic plots indicated no major deviations from these assumptions.

One-Way ANOVA

A **one-way Analysis of Variance (ANOVA)** was conducted to test whether mean task accuracy differs across multitasking groups.

Participants were categorized into three levels:

- Low multitasking
- Moderate multitasking
- High multitasking

ANOVA compares the **between-group variance** in scores to the **within-group variance** to determine whether group differences are statistically significant.

Assumptions of One-Way ANOVA

One-way ANOVA requires the following assumptions:
• **Independence of observations** – Each participant contributes only one observation.
This assumption is satisfied because the dataset consists of independent survey responses.

• **Normality of residuals** – The residuals within each group should be approximately normally distributed.
Residual diagnostics indicated no strong deviations from normality.

• **Homogeneity of variance** – Variance of scores should be similar across groups.
Group variance patterns did not indicate major violations of this assumption.

These diagnostics support the validity of the ANOVA results.

Independent Samples t-Test

An **independent samples t-test** was conducted to compare mean task accuracy between fast and slow task completion groups.

Participants were divided into two groups:

- Fast responders
- Slow responders

The t-test evaluates whether the mean scores of two independent groups differ significantly.

Assumptions of the t-Test

Key assumptions include:

- independence of observations
- approximately normal distribution of scores within groups
- homogeneity of variances

These assumptions were assessed prior to interpreting the test results.

Wilcoxon Rank-Sum Test

Because the distribution of completion times may deviate from normality and the slow responder group contained fewer observations, a **Wilcoxon rank-sum test** was also performed.

The Wilcoxon test is a **non-parametric alternative to the independent samples t-test** and does not assume normality of the underlying data.

This test compares the median ranks of the two groups to determine whether their distributions differ significantly.

Assumptions of the Wilcoxon Rank-Sum Test

The Wilcoxon rank-sum test is a non-parametric alternative to the t-test and relies on fewer assumptions:
• independence of observations
• ordinal or continuous measurement scale
• similar distribution shapes between groups

Because the Wilcoxon test does not require normality, it provides a robust comparison when normality assumptions may be violated.

Behavioural Mechanism Model

Based on the combined statistical findings, a conceptual behavioural mechanism model was developed to explain the relationship between digital multitasking behaviour and cognitive performance.

The proposed mechanism suggests that multitasking behaviour influences task performance indirectly through attentional processes such as notification interruptions and focus level.
The conceptual pathway can be summarized as:

Multitasking Behaviour → Notification Interruptions → Focus Level → Task Accuracy

This framework integrates results from regression, mediation, and clustering analyses to provide a comprehensive behavioural explanation of cognitive performance in digitally intensive study environments.

All statistical analyses were conducted at a significance level of $\alpha = 0.05$.

3.7 Reproducibility and Software Environment

All statistical analyses were conducted using the R programming language. Data preprocessing and storage were performed using MySQL. The R packages used included packages for regression modelling, clustering, and mediation analysis. The use of scripted workflows ensures that all analytical steps are reproducible.

4. Results

4.1 Descriptive Statistics

The final dataset consisted of **139 observations** collected from students who completed an online survey and a short cognitive task assessment. The dataset included measures of digital multitasking behaviour, notification frequency, screen time exposure, attentional focus, and task accuracy.

Descriptive statistics were computed to summarize the central tendencies and variability of the variables included in the analysis.

Variable	Mean	Standard Deviation	Minimum	Maximum
Score	9.71	2.63	2	14
Multitasking Index	1.52	0.59	0	3
Screen Time	2.73	0.89	1	4
Notifications	1.94	1.11	1	4
Focus Level	5.80	2.42	1	10

The average task accuracy score was **9.71 (SD = 2.63)** out of a maximum possible score of 15. Scores ranged from **2 to 14**, indicating moderate performance among participants.

The mean **multitasking index** was **1.52**, suggesting that students generally engaged in moderate levels of digital multitasking behaviour during study activities. Notification frequency averaged **1.94**, while average screen time exposure was **2.73** on the categorical scale used in the survey.

Focus levels varied across participants (Mean = **5.80**, SD = **2.42**), indicating substantial differences in attentional control during the cognitive task.

Overall, the descriptive statistics indicate that the dataset contains meaningful variability across behavioural variables, which allows for further statistical modelling and analysis.

Figure 1 — Histogram of Score Distribution

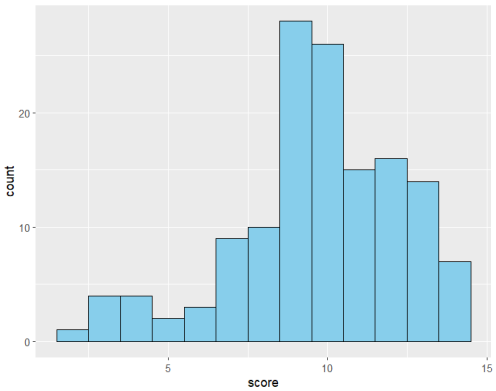

Figure 1: Distribution of cognitive task scores among participants.
The histogram shows that task scores are approximately symmetrically distributed, with most observations falling between **8 and 12**, suggesting that the task had an appropriate level of difficulty without strong ceiling or floor effects.

4.2 Exploratory Data Analysis

Exploratory data analysis (EDA) was conducted to visually examine the relationships between digital multitasking behaviours and cognitive task accuracy before applying statistical models. Scatterplots were used to assess potential associations between the task score and key behavioural variables, including multitasking index, notification frequency, and focus level.

Multitasking and Task Accuracy:
A scatterplot was generated to examine the relationship between the multitasking index and task accuracy scores.

Figure 2: Scatterplot of Multitasking Index vs Task Score.

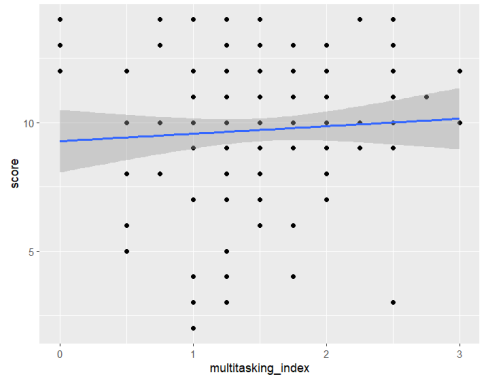

Figure 2: The plot shows that the relationship between multitasking behaviour and task accuracy is relatively weak. Data points are widely dispersed, indicating no strong linear pattern between the multitasking index and cognitive performance.

Completion Time and Task Performance:
A scatterplot was also generated to explore the relationship between task completion time and task accuracy.

Figure 3: Scatterplot of Completion Time vs Task Score.

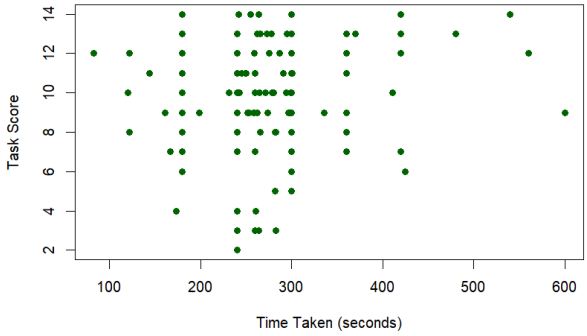

Figure 3: The plot suggests a weak positive trend, indicating that participants who spent slightly more time completing the task tended to achieve marginally higher scores. However, the relationship appears relatively weak.

Interpretation of Exploratory Patterns

Overall, the exploratory analysis suggests that multitasking behaviour does not exhibit a strong visual relationship with task accuracy. Instead, task performance appears to be influenced by multiple behavioural and attentional factors. These initial observations provide motivation for the subsequent correlation and regression analyses.

4.3 Correlation Analysis

To examine the relationships between task accuracy and behavioural variables, **Pearson correlation coefficients** were computed. Correlation analysis provides a preliminary assessment of the strength and direction of associations between variables before applying regression models.

Figure 4: Correlation Matrix

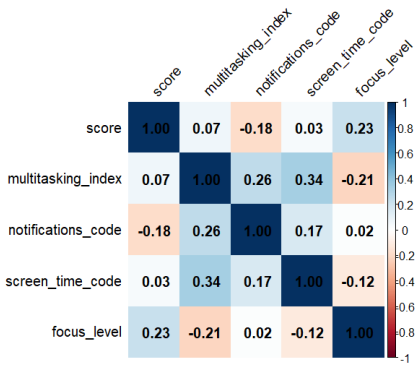

Figure 4: Correlation matrix showing relationships between task score, multitasking behaviour, screen time exposure, notification frequency, and focus level.

Interpretation

The correlation analysis revealed several notable relationships between behavioural variables and cognitive task performance.

Variable Pair	Correlation (r)
Score – Multitasking Index	0.07
Score – Notifications	−0.18
Score – Screen Time	0.03
Score – Focus Level	0.23

The strongest relationship was observed between **focus level and task accuracy (r = 0.23)**, suggesting that students who reported higher attentional focus tended to perform slightly better on the cognitive task.

Notification frequency showed a **weak negative correlation with task score (r = −0.18)**, indicating that more frequent digital interruptions may be associated with lower task accuracy. In contrast, multitasking behaviour and screen time exhibited **very weak correlations with performance**, suggesting that these factors alone may not strongly influence cognitive task outcomes.

Overall, the correlation results suggest that **attentional factors such as focus and digital interruptions may play a more important role in cognitive task performance than multitasking behaviour itself**. These findings provide preliminary evidence that motivates the regression analysis presented in the following section.

4.4 Regression Analysis

To investigate the relationship between digital multitasking behaviour and cognitive task accuracy, a series of **linear regression models** were estimated. The models were constructed progressively to examine how different behavioural variables contribute to the prediction of task accuracy.

Model 1 — Multitasking Behaviour

The first model examined whether multitasking behaviour alone predicts task accuracy.

Model 1:
Score = $\beta_0 + \beta_1$(Multitasking Index) + ε

The regression analysis indicated that multitasking behaviour had **very limited explanatory power** for predicting task accuracy.

The model produced an **R^2 value of approximately 0.004**, meaning that multitasking behaviour explained less than **1% of the variance** in task scores.

Interpretation
This result suggests that multitasking behaviour alone does not significantly predict cognitive task performance. Students who reported higher multitasking levels did not necessarily perform worse on the task.

Model 2 — Adding Screen Time

The second model included screen time exposure as an additional predictor.
Model 2:
Score = $\beta_0 + \beta_1$(Multitasking Index) + β_2(Screen Time) + ε

The inclusion of screen time did **not substantially improve model performance**, and the explanatory power of the model remained very low.

Interpretation
Screen time exposure did not appear to significantly influence cognitive task accuracy in this dataset.

Model 3 — Full Behavioural Model

The third model incorporated additional behavioural variables, including notification frequency and focus level.

Model 3:

$$\text{Score} = \beta_0 + \beta_1(\text{Multitasking Index}) + \beta_2(\text{Screen Time}) + \beta_3(\text{Notifications}) + \beta_4(\text{Focus Level}) + \varepsilon$$

This model showed a **notable improvement in explanatory power**, with an **R^2 value of approximately 0.118**.
Two predictors emerged as particularly important:

- **Notification frequency** showed a **negative association** with task accuracy.
- **Focus level** showed a **positive association** with task performance.

Interpretation
Students who experienced more frequent digital notifications tended to perform slightly worse on the cognitive task, while those reporting higher levels of attentional focus achieved higher scores.

Model 4 — Interaction Model

To investigate whether multitasking becomes more disruptive in the presence of digital interruptions, an interaction term between multitasking behaviour and notification frequency was included.

Model 4:

$$\text{Score} = \beta_0 + \beta_1(\text{Multitasking}) + \beta_2(\text{Notifications}) + \beta_3(\text{Multitasking} \times \text{Notifications}) + \beta_4(\text{Screen Time}) + \beta_5(\text{Focus Level}) + \varepsilon$$

The interaction model produced an **R^2 value of approximately 0.166**, representing the strongest model among those tested.

Interpretation
The results suggest that the relationship between multitasking behaviour and task accuracy depends on notification frequency. Multitasking behaviour appears to become **more disruptive when students experience frequent digital interruptions**.

<u>Model Comparison:</u>

A comparison of the regression models shows that explanatory power increases as additional behavioural variables are included.

Model	Variables Included	R^2
Model 1	Multitasking	0.004
Model 2	Multitasking + Screen Time	~0.004
Model 3	Multitasking + Screen Time + Notifications + Focus	0.118
Model 4	Multitasking × Notifications Interaction	0.166

These results indicate that **attentional factors such as notification frequency and focus level play a more important role in predicting task accuracy than multitasking behaviour alone**.

4.5 Regression Diagnostics

Before interpreting the regression results, several diagnostic tests were conducted to verify that the assumptions of linear regression were reasonably satisfied. These diagnostic checks included the examination of residual distributions, normality, influential observations, and multicollinearity.

Residual Distribution

A histogram of the regression residuals was examined to assess whether the residuals were approximately normally distributed.

Figure 5: Histogram of regression residuals.

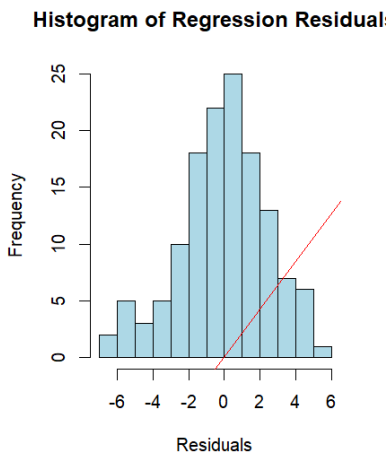

Figure 5: The histogram shows that the residuals are roughly symmetric and centered around zero. This suggests that the normality assumption of the regression model is reasonably satisfied.

Residual vs Fitted Plot

A residual versus fitted values plot was used to evaluate the assumption of **homoscedasticity**, which requires that the variance of residuals remains constant across predicted values.

Figure 6: Residual vs Fitted plot

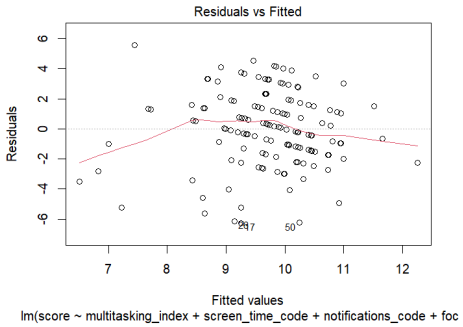

Figure 6: The residuals appear randomly scattered around the horizontal axis without clear systematic patterns. This indicates that the model does not suffer from strong heteroscedasticity and that the assumption of constant variance is reasonably satisfied.

Normal Q–Q Plot

A Q–Q plot was generated to further assess the normality of the regression residuals.

Figure 7: Normal Q–Q plot of regression residuals.

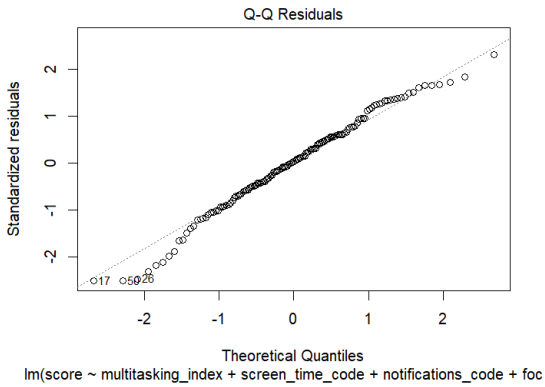

Figure 7: Most points lie close to the diagonal reference line, indicating that the residuals follow an approximately normal distribution. Minor deviations at the extremes are expected in behavioural datasets and do not substantially affect the validity of the regression model.

Influential Observations

Potential influential observations were evaluated using **Cook's Distance**.

Figure 8: Cook's Distance plot.

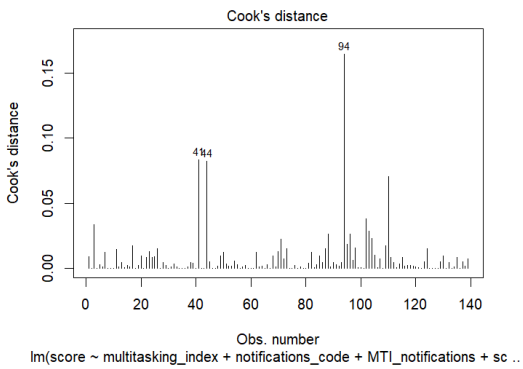

Figure 8: All Cook's distance values were well below the commonly used threshold of 1, suggesting that no individual observation exerted excessive influence on the regression results.

Multicollinearity Assessment

Multicollinearity among predictors was assessed using the **Variance Inflation Factor (VIF)**.

Predictor	VIF
Multitasking Index	4.23
Notifications	7.72
Interaction Term	12.99
Screen Time	1.15
Focus Level	1.07

The higher VIF for the interaction term is expected because interaction variables are mathematically correlated with their component variables.

Most predictors exhibited VIF values close to 1, indicating low multicollinearity. Although the interaction term showed a higher VIF value, this is expected because interaction variables are mathematically related to their component variables.

Overall, the regression diagnostics indicate that the key assumptions of linear regression were reasonably satisfied. Therefore, the regression results can be interpreted with confidence.

4.6 Group Comparisons

To further examine whether behavioural differences influence task accuracy, group comparison analyses were conducted. These analyses evaluated whether task performance differed across multitasking levels and task completion speed.

Multitasking Level Comparison

Participants were categorized into three multitasking groups based on their multitasking index values:

- **Low multitasking**
- **Moderate multitasking**
- **High multitasking**

A **one-way Analysis of Variance (ANOVA)** was performed to determine whether mean task accuracy differed across these groups.

ANOVA Results

Source	Df	Sum Sq	Mean Sq	F	p-value
Multitasking Group	2	8.1	4.037	0.579	0.562
Residuals	136	948.4	6.974		

Since the **p-value (0.562)** is greater than the significance level of **0.05**, the null hypothesis cannot be rejected.

Interpretation:

The ANOVA results indicate that there is **no statistically significant difference in task accuracy across the multitasking groups**. Students with low, moderate, and high multitasking levels achieved similar average task scores.

This finding supports earlier results from the correlation and regression analyses, which suggested that multitasking behaviour alone does not strongly influence cognitive task performance.

Completion Time Comparison

To examine whether task completion speed influenced task accuracy, participants were divided into two groups:

- **Fast responders** (≤ 7 minutes)
- **Slow responders** (> 7 minutes)

A **Welch two-sample t-test** was conducted to compare mean task scores between these groups.

t-test Results
$t = -0.758$
$p = 0.489$

The results indicate **no statistically significant difference** in task accuracy between fast and slow responders.

Non-Parametric Comparison

Because the slow responder group contained a relatively small number of observations, a **Wilcoxon rank-sum test** was also conducted as a non-parametric alternative.

Wilcoxon Test Results
$W = 245.5$
$p = 0.353$

The Wilcoxon test also indicated **no statistically significant difference** between the two groups.

Interpretation

Both parametric and non-parametric tests consistently show that **task completion speed does not significantly influence cognitive task accuracy** in this dataset. Although slower participants tended to have slightly higher scores on average, the difference was not statistically significant.
These findings further reinforce the overall conclusion that behavioural factors such as **notification interruptions and attentional focus may play a more important role in task performance than multitasking behaviour or task completion speed alone**.

4.7 Behavioural Clustering Analysis

To explore patterns in students' digital study behaviours, **K-means clustering** was applied using four behavioural variables:

- Multitasking Index
- Notification Frequency
- Screen Time
- Focus Level

Before clustering, all variables were **standardized** to ensure that differences in measurement scales did not influence the clustering results.

The **elbow method** was used to determine the optimal number of clusters. The results suggested that a **three-cluster solution** provided a suitable balance between model simplicity and explanatory power.

Cluster Characteristics:

The clustering analysis grouped participants into three behavioural profiles.

Cluster	Mean Task Score
Cluster 1	9.83
Cluster 2	9.35
Cluster 3	9.79

The clusters represent different combinations of multitasking behaviour, notification exposure, screen time, and focus levels.

Cluster Interpretation

Although distinct behavioural patterns were identified among students, the **average task scores across clusters were relatively similar**. This suggests that behavioural grouping alone does not strongly differentiate cognitive performance.

In other words, students may exhibit different patterns of digital study behaviour, but these patterns do not necessarily lead to large differences in task accuracy.

These findings are consistent with the earlier regression and correlation results, which indicated that multitasking behaviour alone does not strongly predict cognitive task performance. While students demonstrate different patterns of digital study behaviour, these patterns do not appear to produce substantial differences in task accuracy.

Instead, the results of the clustering analysis suggest that **specific attentional factors**, particularly notification frequency and self-reported focus level, may play a more important role in determining task performance than multitasking behaviour itself.

Overall, the clustering results reinforce the broader pattern observed throughout the analysis: **digital interruptions and attentional focus appear to have stronger associations with task accuracy than multitasking behaviour alone**.

4.8 Relative Importance Analysis

To evaluate the relative contribution of each predictor to the explained variance in task accuracy, a **relative importance analysis** was conducted using the LMG method. This approach partitions the total coefficient of determination (R^2) across predictors, allowing comparison of their individual contributions to the model.

The analysis revealed that some predictors contributed substantially more to the explained variance in task accuracy than others.

Predictor	Contribution to R^2
Notifications	38.2%
Focus Level	35.3%
Interaction (Multitasking × Notifications)	17.6%
Multitasking Index	7.9%
Screen Time	0.9%

The results indicate that **notification frequency accounts for the largest share of explained variance**, followed closely by **focus level**. The interaction between multitasking behaviour and notification frequency also contributed meaningfully to the model.

In contrast, multitasking behaviour alone contributed relatively little to the prediction of task accuracy, while screen time exposure had a negligible contribution.

These findings suggest that **digital interruptions and attentional focus are the most important determinants of cognitive task performance in this dataset**.

4.9 Mediation Analysis

To further examine the behavioural mechanisms underlying the relationship between multitasking behaviour and task accuracy, **mediation analyses** were conducted.

Two mediation models were tested:

1. Multitasking → Focus Level → Task Score
2. Multitasking → Notification Frequency → Task Score

Mediation Model 1: Focus as a Mediator

The first model examined whether **focus level mediates the relationship between multitasking behaviour and task accuracy**.

The mediation analysis revealed a **significant indirect effect** of multitasking behaviour on task accuracy through focus level.

Effect	Estimate	p-value
ACME (Indirect Effect)	−0.241	0.0316
ADE (Direct Effect)	0.560	0.158
Total Effect	0.319	0.420

The significant indirect effect indicates that higher multitasking behaviour is associated with **reduced attentional focus**, which in turn lowers task accuracy.

However, the **direct effect of multitasking behaviour on task accuracy was not statistically significant**, suggesting that the relationship primarily operates through attentional mechanisms.

Mediation Model 2: Notifications as a Mediator

A second mediation model examined whether **notification frequency mediates the relationship between multitasking behaviour and task accuracy**.

Effect	Estimate	p-value
ACME (Indirect Effect)	−0.259	0.0292
ADE (Direct Effect)	0.579	0.157
Total Effect	0.319	0.430

The results indicate a **significant indirect effect through notification frequency**, suggesting that multitasking behaviour is associated with increased digital interruptions, which subsequently reduce task accuracy.

Interpretation

Together, the mediation analyses suggest that multitasking behaviour influences task performance **indirectly through attentional pathways**, rather than through a direct effect.

Specifically, higher multitasking behaviour is associated with:

- increased digital notifications
- reduced attentional focus

Both of which contribute to **lower cognitive task accuracy**.

4.10 Key Findings Summary

The results of the statistical analyses reveal several important findings regarding the relationship between digital multitasking behaviour and cognitive task performance.

Analysis Method	Main Finding	Interpretation
Descriptive Statistics	Mean score ≈ 9.7	Students showed moderate task performance
Correlation Analysis	Focus positively correlated with score ($r \approx 0.23$)	Higher focus associated with better performance
Correlation Analysis	Notifications negatively correlated with score ($r \approx -0.18$)	Digital interruptions may reduce task accuracy
Regression Analysis	Notifications negative, focus positive	Attentional factors influence task performance
Interaction Model	Multitasking × Notifications significant	Multitasking becomes more disruptive when interruptions increase
Relative Importance	Notifications (38%) and focus (35%) dominate	Interruptions and attention are key predictors
Clustering Analysis	Behaviour clusters identified	Behaviour patterns exist but score differences are small
ANOVA	No difference across multitasking groups ($p = 0.562$)	Multitasking level alone does not strongly affect performance
Time Comparison	No significant difference ($p > 0.05$)	Completion speed does not strongly influence accuracy
Mediation Analysis	Significant indirect effects	Multitasking affects performance through attentional mechanisms

Overall, the results consistently indicate that **digital interruptions and attentional focus play a stronger role in determining task accuracy than multitasking behaviour alone**.

5. Discussion

The present study investigated the relationship between **digital multitasking behaviour and cognitive task accuracy among students**. With the increasing integration of digital technologies in educational environments, students frequently engage with multiple digital streams simultaneously, including smartphones, laptops, notifications, and background media. Understanding how such behaviours influence cognitive performance is therefore increasingly important.

The results of this study suggest that **digital interruptions and attentional focus play a more substantial role in task accuracy than multitasking behaviour alone**. While multitasking behaviour was initially hypothesized to negatively affect cognitive performance, the findings indicate that multitasking itself showed only a weak independent relationship with task accuracy. Instead, variables related to **attention and digital interruptions**, particularly notification frequency and focus level, were more strongly associated with variations in performance.

The correlation analysis revealed that **focus level exhibited a positive relationship with task accuracy**, indicating that students who reported higher attentional focus tended to achieve better scores on the cognitive task. In contrast, notification frequency showed a **negative relationship with task performance**, suggesting that frequent digital interruptions may disrupt sustained attention during cognitive tasks. These findings highlight the importance of attentional control when students interact with digital learning environments.

Regression analysis further supported these observations. Models that included only multitasking behaviour explained very little variation in task accuracy. However, when attentional variables such as notification frequency and focus level were introduced, the explanatory power of the model increased considerably. The interaction model also suggested that multitasking behaviour may become more disruptive when combined with frequent digital interruptions. This indicates that the impact of multitasking behaviour may depend on the broader context of digital interruptions experienced by students.

These findings align with prior research examining the cognitive consequences of media multitasking. For example, **Uncapher, Thieu, and Wagner (2016)** reported that individuals who frequently engage in media multitasking often exhibit **reduced working memory performance and increased susceptibility to distraction**. Their findings suggest that heavy media multitaskers may have difficulty filtering irrelevant information, which can interfere with goal-directed cognitive processing.

The results of the present study support a similar attentional explanation. Although multitasking behaviour itself was not found to be a strong independent predictor of task accuracy, factors related to **digital interruptions and attentional control** appear to influence cognitive performance. This suggests that the cognitive challenges associated with digital multitasking may arise primarily from **attentional disruption rather than multitasking behaviour alone**.

Importantly, the present study extends previous research by examining multitasking behaviour in a **realistic academic context** rather than controlled laboratory tasks. Earlier studies often focused on memory tasks conducted under experimental conditions, whereas the current research used **primary student data and a cognitive reasoning test** to measure task performance. This approach allows for a more realistic assessment of how digital multitasking behaviours operate in everyday study environments.

Additional analyses further reinforced these conclusions. The clustering analysis identified different patterns of digital study behaviour among students, but these behavioural groups did not show large differences in average task accuracy. Similarly, group comparisons using ANOVA indicated that multitasking levels alone did not significantly differentiate performance outcomes. These findings suggest that while students may adopt different digital study habits, these behaviours do not necessarily translate into substantial differences in cognitive task performance.

The mediation analysis provided further insight into the behavioural mechanisms underlying these relationships. The results indicated that multitasking behaviour may influence task performance **indirectly through attentional mechanisms**, particularly through reduced focus and increased digital interruptions. This supports the idea that multitasking behaviour may contribute to cognitive performance differences by influencing how attention is allocated during tasks.

Taken together, the findings of this study suggest that **digital interruptions and attentional focus are key determinants of task accuracy in digital learning environments**. While multitasking behaviour is often assumed to impair performance directly, the results indicate that the effects of multitasking may be mediated by attentional processes and the frequency of digital interruptions.

These insights have important implications for understanding how students interact with modern digital learning environments. As digital technologies continue to play a central role in education, identifying factors that support sustained attention and reduce unnecessary interruptions may help improve cognitive performance during academic tasks.

6. Conclusion

This study examined the relationship between **digital multitasking behaviour and cognitive task accuracy among students**. Using a dataset of 139 student responses, the research analyzed how behavioural factors such as multitasking, notification frequency, screen time exposure, and attentional focus relate to performance on a cognitive task.

The findings indicate that **multitasking behaviour alone does not strongly predict task accuracy**. Instead, attentional factors such as **notification frequency and focus level** were found to play a more significant role in influencing performance. Students who reported higher levels of attentional focus tended to achieve better task scores, while those experiencing frequent digital interruptions showed slightly lower levels of task accuracy.

Regression analysis demonstrated that models incorporating attentional variables explained substantially more variation in task performance than models relying on multitasking behaviour alone. Additionally, the interaction analysis suggested that multitasking behaviour may become more disruptive when students experience frequent digital interruptions.

These findings contribute to the growing literature on digital multitasking and cognitive performance. Consistent with previous research on media multitasking and attention, the results suggest that **attentional disruption rather than multitasking behaviour itself may be the primary mechanism influencing cognitive performance**. The study therefore highlights the importance of managing digital interruptions in modern learning environments.

Overall, the results suggest that improving attentional focus and reducing unnecessary digital interruptions may support better cognitive performance in digitally mediated study settings.

The findings of this study have practical implications for digital learning environments.
Reducing unnecessary digital notifications during study sessions may help students maintain attentional focus and improve cognitive task performance.

Educational institutions may also consider encouraging structured digital study practices that minimize interruptions and support sustained attention.

Practical Implications

- Students should minimize digital notifications during study sessions to maintain attentional focus.
- Educational institutions may encourage distraction-free study environments and structured digital usage practices.
- Digital learning platforms may consider implementing “focus modes” that temporarily limit notifications during academic tasks.

7. Limitations

Although this study provides insights into the relationship between digital multitasking behaviour and task accuracy, several limitations should be acknowledged.

First, the behavioural variables used in the study were **self-reported**, which may introduce measurement bias. Participants may not always accurately estimate their multitasking behaviour or digital device usage.

Second, the study used a **cross-sectional research design**, which limits the ability to draw causal conclusions. While associations between variables were identified, it cannot be definitively concluded that multitasking behaviour directly causes changes in task accuracy.

Third, the sample size was limited to **student participants**, which may restrict the generalizability of the findings to broader populations. Different age groups or professional environments may exhibit different patterns of digital multitasking behaviour.

Finally, the cognitive task used in the study represents only one form of cognitive performance. Other aspects of cognition, such as memory, attention switching, or problem-solving ability, may be affected differently by digital multitasking behaviours.

8. Future Research

Future research could extend the findings of this study in several ways.

First, studies with **larger and more diverse datasets** could help confirm whether the observed relationships between digital interruptions, attentional focus, and task accuracy generalize across different populations.

Second, **experimental or longitudinal research designs** could provide stronger evidence regarding the causal relationship between digital multitasking behaviours and cognitive performance.

Third, future studies could incorporate **objective behavioural measurements**, such as device usage logs or screen tracking data, to complement self-reported measures of multitasking behaviour.

Finally, additional cognitive tasks could be included to examine how digital multitasking affects **different dimensions of cognitive performance**, such as working memory, attention switching, and problem-solving ability.

Such research would further improve our understanding of how digital multitasking behaviour influences learning and cognitive functioning in increasingly technology-driven environments.

9. Data Availability Statement

The dataset used in this study was collected through an online survey conducted using Google Forms. The processed dataset used for statistical analysis contains anonymized responses from 139 participants.

Due to privacy considerations, individual-level responses are not publicly distributed. However, aggregated data and derived variables used in the statistical analyses may be made available upon reasonable request for academic purposes.

10. Software and Reproducibility Statement

All statistical analyses were conducted using the R programming language. Data preprocessing and database management were performed using MySQL. The analysis pipeline included procedures for data cleaning, exploratory data analysis, regression modelling, clustering analysis, and mediation analysis.

All statistical scripts were implemented using reproducible workflows to ensure that the analysis can be replicated using the same dataset and computational environment.

11. References

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12. Declaration

I hereby declare that the research project titled **"Digital Multitasking and Cognitive Task Accuracy"** is my original work carried out as part of the MSc Statistics programme at Bhavan's College.

The conceptual framework and inspiration for the study were partly informed by existing literature on digital multitasking and cognitive performance, particularly the study by **Uncapher, Thieu, and Wagner (2016) on media multitasking and memory.** However, the present research uses independently collected survey data and a separate analytical framework designed specifically for this project.

All sources of information, theoretical ideas, and prior research used in this study have been properly acknowledged through citations and references.

The data collection, statistical analysis, interpretation of results, and writing of this report were conducted independently by the author.

Shaikh Maryam

13. Appendix

Appendix A: Survey Questionnaire

Digital Multitasking and Task Accuracy Study

This questionnaire was designed to collect information about students' digital study behaviour and cognitive task performance. The survey formed part of an academic research project investigating the relationship between digital multitasking and task accuracy.

Participants were informed that their responses would remain anonymous and would be used only for research purposes.

Section 1: Demographic Information

1. Age Group

- Below 20
- 20–22
- 23–25
- Above 25

2. Gender

- Male
- Female
- Prefer not to say

Section 2: Digital Study Behaviour

3. While studying, how often do you switch between apps or browser tabs?

- Never
- Rarely
- Sometimes
- Often
- Always

4. How frequently do you check your phone while studying?

- Never
- Rarely

- Sometimes
- Often
- Always

5. On average, how many notifications do you receive during one study session?

- 0–5
- 6–10
- 11–20
- More than 20

6. Do you usually use more than one device at the same time while studying?

- Yes
- No

7. Do you use any background media while studying?

- None
- Music only
- Videos/Reels
- Both music and videos

8. How often do you use AI-based tools (e.g., ChatGPT, Gemini, etc.) while studying or completing academic tasks?

- Never
- Rarely
- Sometimes
- Often
- Always

9. How focused do you generally feel during your study sessions?

Scale:
1 = Not focused at all
10 = Extremely focused

Participants selected a value from **1 to 10** indicating their perceived level of focus.

10. Average daily screen time

- Below 2 hours
- 2–4 hours

- 4–6 hours
- Above 6 hours

11. Difficulty staying focused while studying

- Never
- Rarely
- Sometimes
- Often
- Always

Section 3: Cognitive Task Assessment

Participants completed a **15-question reasoning test** designed to evaluate cognitive task accuracy. The test included questions related to logical reasoning, pattern recognition, and attention to detail.

Each correct answer contributed **one point** toward the participant's final task accuracy score. The total possible score ranged from **0 to 15**, representing the number of correctly answered questions.

Appendix B: Cognitive Task Assessment

The second part of the survey consisted of a short cognitive reasoning assessment designed to measure task accuracy. The test included fifteen questions intended to evaluate logical reasoning, pattern recognition, and attention to detail.

Participants were instructed to answer each question carefully. Each correct answer contributed one point toward the final task accuracy score. The total possible score ranged from **0 to 15**.

The cognitive task included questions of the following types:

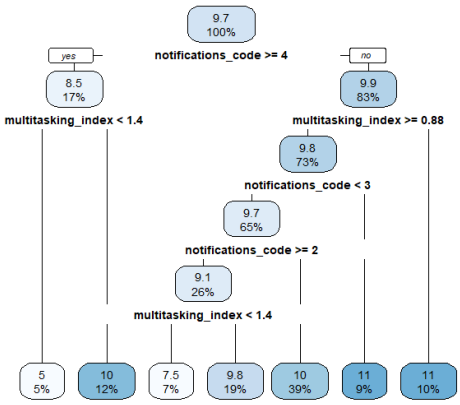
1. **Number Pattern Recognition**
   Participants identified the missing value in a numerical sequence.
2. **Logical Reasoning Problems**
   Participants solved short logical deduction questions requiring reasoning based on given statements.
3. **Symbol and Pattern Matching**
   Participants selected the correct symbol or pattern that completed a sequence.
4. **Visual Pattern Completion**
   Participants determined the next item in a visual pattern.
5. **Analogy Questions**
   Participants identified relationships between pairs of words or concepts.
6. **Basic Mathematical Reasoning**
   Simple arithmetic or logical calculation problems.
7. **Series Identification**
   Participants identified the correct continuation of a numerical or symbolic series.
8. **Attention-Based Questions**
   Questions designed to test careful reading and attention to detail.
9. **Logical Elimination Problems**
   Participants selected the correct option by eliminating inconsistent alternatives.
10. **Reasoning Puzzles**
    Short puzzles requiring step-by-step logical thinking.

Each participant completed all fifteen questions within the survey form. The total number of correct responses was recorded as the **task accuracy score**, which served as the primary dependent variable in the statistical analysis.

The responses were automatically recorded through Google Forms and exported into the dataset used for statistical analysis.

Appendix C: EDA Graphs

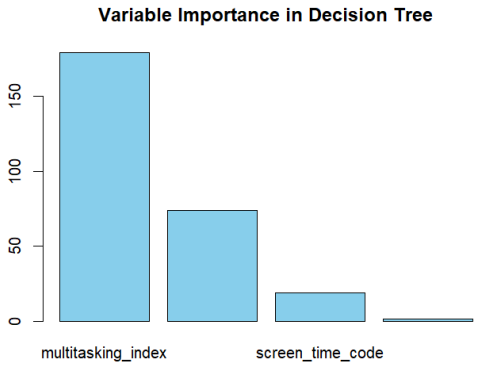
Figure 1: Decision Tree Model for Predicting Task Accuracy



Interpretation

The decision tree indicates that **notification frequency and multitasking behaviour** are key variables influencing task accuracy. Higher notifications combined with higher multitasking levels are associated with **lower predicted scores**, suggesting that digital interruptions may negatively affect cognitive performance.

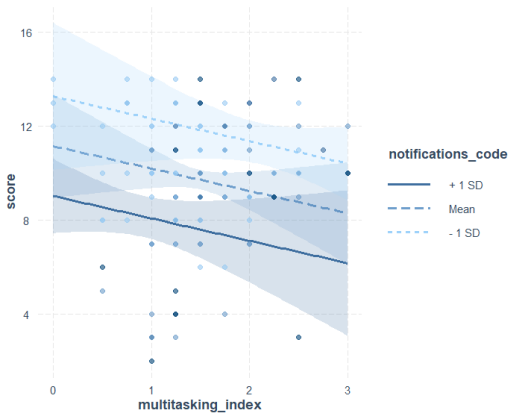
Figure 2: Variable Importance in Decision Tree



Interpretation

The variable importance plot shows that **multitasking behaviour is the most influential predictor**, followed by notification frequency, while screen time contributes relatively little to predicting task accuracy.

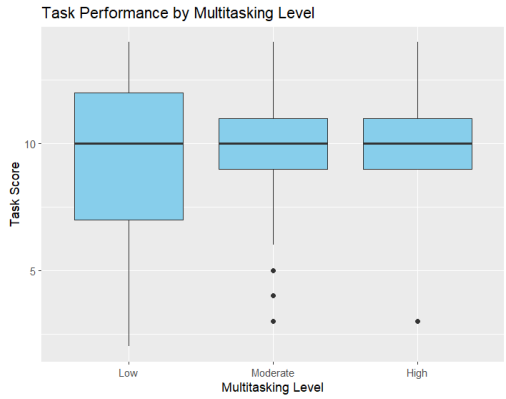
Figure 3: Interaction Between Multitasking and Notifications



Interpretation

The plot shows a declining relationship between multitasking behaviour and task accuracy. The negative effect becomes stronger at higher levels of notification frequency, indicating that notifications may amplify the impact of multitasking on performance.

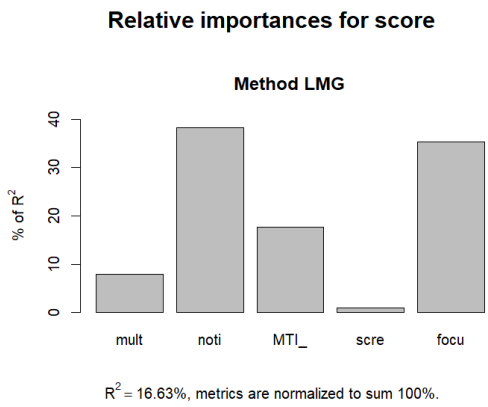
Figure 4: Task Performance by Multitasking Level



Interpretation

Task scores appear relatively similar across multitasking groups, with overlapping distributions. This suggests that **differences in mean task accuracy across multitasking levels are not substantial**.

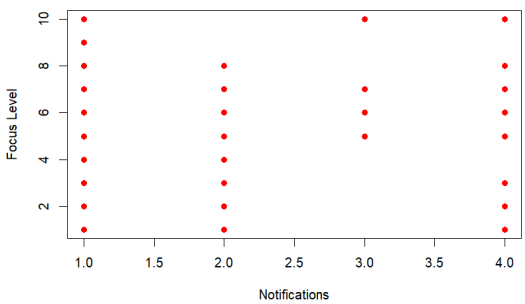
Figure 5: Relative Importance of Predictors



Interpretation

Relative importance analysis indicates that **notification frequency and focus level explain the largest proportion of variance in task accuracy**, while screen time contributes minimally.

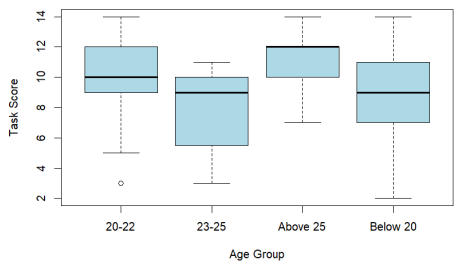
Figure 6: Notifications vs Focus Level



Interpretation

The scatterplot suggests that **higher notification levels may be associated with lower focus levels**, indicating a possible relationship between digital interruptions and reduced attentional focus.

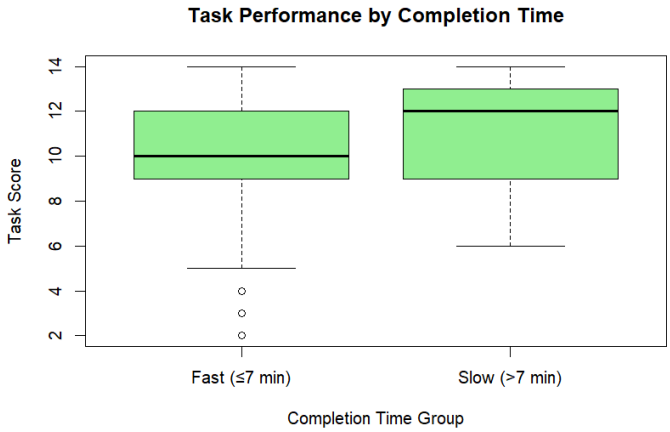
Figure 7: Task Accuracy by Age Group



Interpretation

Task scores appear relatively consistent across age groups, indicating that **age does not strongly influence task accuracy within the sample**.

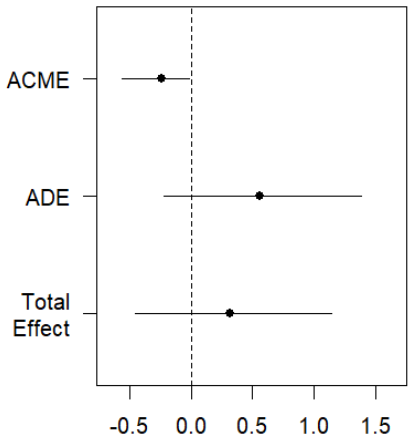
Figure 8: Task Performance by Completion Time



Interpretation

Participants who spent more time completing the task show slightly higher median scores, although the distributions overlap. This suggests **no strong difference in accuracy between fast and slow responders**.

Figure 9: Mediation Analysis



Interpretation

The mediation plot indicates that **multitasking behaviour may influence task accuracy indirectly through intermediate variables such as focus level or notifications**, suggesting the presence of indirect behavioural pathways.